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Paleomagnetic Study of Devonian Rocks From Ste. Cécile–St. Sébastien Region, Quebec Appalachians

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Three Devonian rock units were collected for paleomagnetic study from the Ste. Cécile–St. Sébastien region located in the Gaspé–Connecticut Valley synclinorium of Québec Appalachians: 15 sites (48 samples) from the metasediments (subgreenschist facies-siltstones, sandstones) of the Compton Formation (early Middle to late Early Devonian age based on fossil plants) which were tightly folded during Middle Devonian Acadian orogeny; 14 sites (61 samples) from the Ste. Cécile–St. Sébastien granitic body (362 m.y., K-Ar age) which intruded the metasediments; 14 sites (62 samples) from the hornfels created subsequent to the emplacement of the intrusive. Detailed thermal cleaning of the metasediments yielded a mean remanence direction characterized by linear vector trajectories, after tilt correction, at D = 209°, I = +20°, k = 30, α_{95} = 7.2° with unblocking temperature spectrum, 300°–550°C, carried probably by magnetite. The predeformational origin of this remanence is tested at 99% confidence level as well as antipodal normal and reverse polarity directions. The corresponding paleomagnetic pole, MS, falls at 28°N, 77°E (*dp*, *dm* = 6°, 11°) in agreement with the recently reported poles which are mostly Early Devonian in age and derived from the southeastern margin of the presumed proto-Atlantic ocean. AF demagnetization of granites resulted in a univectorial mean remanence at: (1) $D = 332°$, $I = -39°$; $k = 31$ for $N = 5$ sites in fine-grained granite (FG) unit; (2) $D = 313°$, $I = -25°$; $k = 19$ for $N = 9$ sites in medium-grained granite (MG) unit; and (3) $D = 319°$, $I = -30°$, $k = 17$ for all the total ($N = 14$) granite (G) sites. AF demagnetization of hornfels revealed a post-intrusive mean remanence direction, HF, at $D = 346°$, $I = -22°$, $k = 8$ for $N = 14$ sites. The individual site-mean remanence directions (with coercivity spectra: 10–60 mT) from both granites and hornfels suffer from a large declination spread of about 60° but the confidence ovals for the FG and HF remanences overlap. This indicates that the FG and HF remanences are indistinguishable from each other and therefore are proximal in time and probably originated during Middle to Late Devonian period. Interpretation of this kind is speculative in the absence of a convincing evidence for a two (FG younger than MG) or three phase intrusive history of the pluton. Since the pole positions of G (18°N, 150°E), MG (18°N, 158°E), FG (18°N, 137°E) and HF (32°N, 125°E) spread over 40° in longitude, these data cannot confirm or negate the possibility of a Mid-Paleozoic transcurrent motion proposed for northern Appalachians. We regard only the pole, MS, with probable late Early or early Middle Devonian age as reliable for the apparent polar path construction.

INTRODUCTION

Devonian paleomagnetic results obtained from northern Appalachians have been interpreted in terms of a regional transcurrent motion during mid-Paleozoic time [*Morris*, 1976; *Kent and Opdyke*, 1978, 1979, 1980; *Van der Voo et al.*, 1979]. Other alternatives such as rapid apparent polar wander during Siluro-Devonian time [*Lapointe*, 1979; *Roy et al.*, 1979], and possible presence of a southerly shear zone adjoining the southeastern margin of Iapetus [*Rao et al.*, 1981] are also explored. *Roy and Anderson* [1981] have discussed different alternatives to interpret the data and the difficulties in establishing both the rock and remanence ages. After this paper was submitted for publication, *Roy* [1982] proposed the possibility of a systematic age progression (from Ordovician to Devonian) of paleolatitudes for northern Appalachians based on the simplest possible polar path approach while *Kent and Opdyke* [1982] have disagreed with such a displacement history.

In this study we report paleomagnetic results from three rock units: (1) Early Devonian metasediments of Compton Formation, (2) a granitic stock which intruded the metasediments some time in Middle Devonian, and (3) its associated hornfels. These are located in a Siluro-Devonian stratigraphic zone known as Gaspé–Connecticut Valley synclinorium (Figures 1*a*, 1*b*) in the southern part of Quebec Appalachians [*St. Julien and Hubert*, 1975; *Schenk*, 1978]. The nature of contact relationship of this zone with respect to the allochthonous Cambro-Ordovician rocks locally known as Internal domain (Figure 1*b*) of Québec Appalachians [*St. Julien and Hubert*, 1975] is poorly understood. Interpretation of a single reflection profile in the study area passing through Lac Drolet (Figure 1*c*) region suggest a fault contact relationship to the northwest and the Siluro-Devonian rock sequences of the synclinorium as thrusted and uplifted blocks [*Séguin*, 1982, Figure 20]. Recently reported seismic results of this region are also in accordance with structural studies [*St. Julien and Hubert*, 1975] suggesting the possible paraautochthonous-allochthonous nature of the lithotectonic belt.

GEOLOGIC SETTING AND AGE RELATIONS

The present study area lies within the Ste. Cécile and St. Sébastien region, north of Mégantic (Figure 1*b*) the geology of which is described in more detail elsewhere [*Kelly*, 1975]. The presence of thick sequences of terrigenous rocks, carbonates and volcanic rocks in this belt suggest that the sedimentary environment changed from marine to continental toward the end of Early Devonian. The sedimentary rocks of the Compton Formation exposed in this region are constituted predominantly of siltstone and sandstone se-

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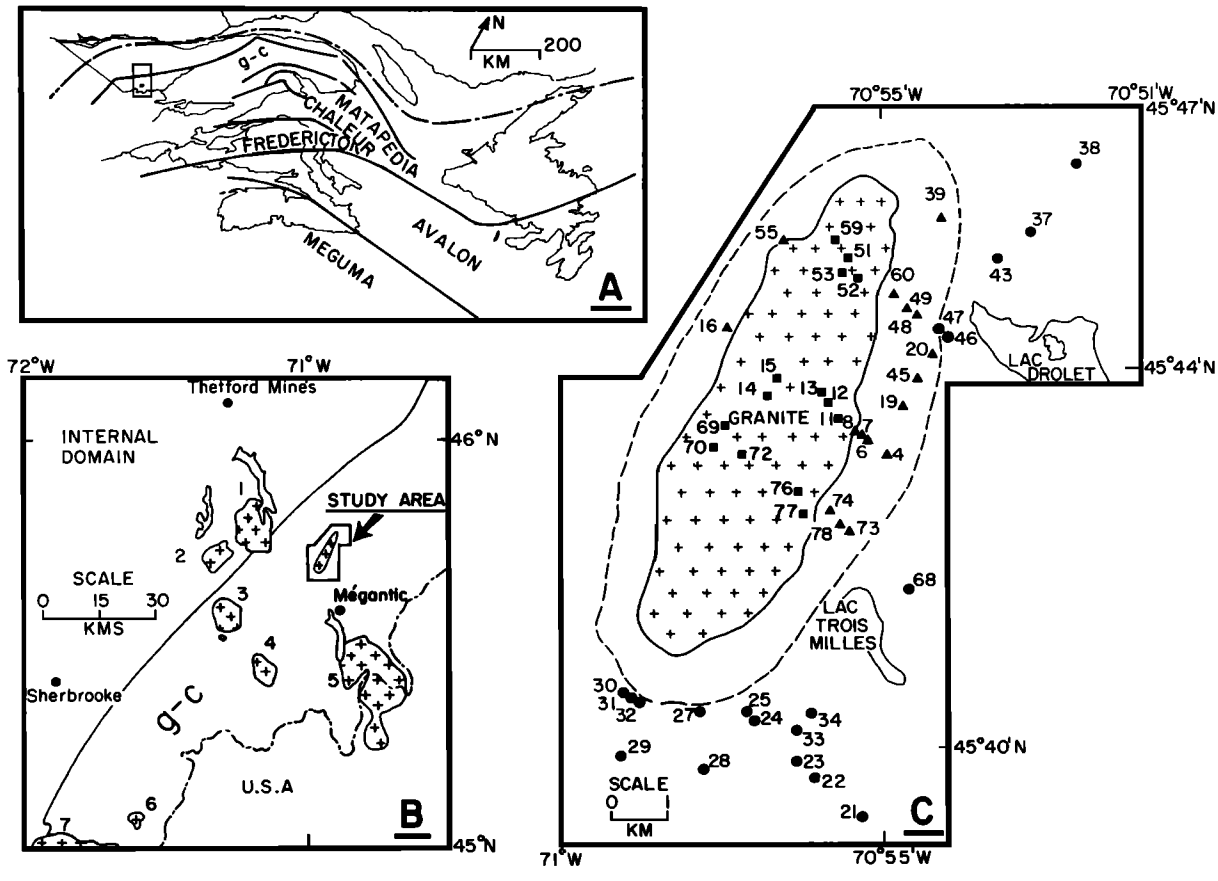


Fig. 1. (*a*) Middle Paleozoic (Siluro-Devonian) tectono-stratigraphic zones in the Canadian Appalachians [*Schenk*, 1978]. A rectangular insert shown in the g-c (Gaspé–Connecticut Valley synclinorium) zone corresponds to the simplified regional geology map in Figure 1*b*. (*b*) Simplified geologic map of a part of southwestern Québec and adjacent New England (U.S.A.) regions [*Cady*, 1969]. The granitic intrusives in this region are numbered (1) Winslow, (2) Mount Aylmer, (3) Scotstown, (4) Mount Megantic, (5) Spider Lake, (6) Mount Hereford, and (7) Stanhope. The location of the Ste. Cécile–St. Sébastien granite and the adjacent area of the present study shown in Figure 1*c* are identified. (*c*) Simplified geologic map of the Ste. Cécile–St. Sébastien region [*Kelly*, 1975] along with the location of the paleomagnetic sampling sites in granites (solid squares), hornfels (solid triangles) and metasediments of the Compton Formation (solid circles). Approximate extent of the geologically inferred contact aureole around the Ste. Cécile–St. Sébastien granite is marked with broken lines.

quences grading into each other. Intercalated bands of slate are also present in the upper parts of the graded beds. The sediments strike N20°E to N70°E and dip steeply (>60°) either to the northwest or to the southeast. They have been weakly metamorphosed and are deformed into tight asymmetric folds with gently-plunging axes.

At the time of Acadian orogeny the metasedimentary unit was later intruded by a large elliptical granitic stock (area of exposure about 36 km^2) locally known as Ste. Cécile–St. Sébastien granite which has created a contact metamorphic aureole of about 1 km in width (Figure 1*c*). The Compton metasediments were subsequently altered to cordierite-andalusite hornfels within this aureole. The intrusion of the stock has also modified appreciably the sedimentary rocks near its periphery by overturning the folds. However, about 2 km eastward from the contact, the folds were not overturned and regional folding seems to be unaffected. *Kelly* [1975] subdivided the granitic stock into medium-grained, fine-grained and aplitic type granites.

Based on the stratigraphic relationships, the Compton metasediments are the oldest in the study area and are overlain unconformably by Cenozoic till and alluvial deposits. The estimated thickness of the Compton sediments vary from 1000 m (Mégantic area) to 2500 m (Sherbrooke area). Preliminary identification of fossil plants discovered in the study area places the age of the Compton Formation between Early and early Middle Devonian [*Kelly*, 1975]. The regional distribution of lithofacies and fossils in the Devonian sediments of the northern Appalachians led *Boucot* [1968, pp. 90–91] to assign Onondaga (Eiefelian or early Middle Devonian) age to the Compton Formation. *Lowdon* [1960] obtained a single K-Ar date of about 362 m.y. from the Ste. Cécile–St. Sébastien granite as an age of intrusion. Similar Middle-Upper Devonian ages of intrusion (360–365 m.y.) for other granitic stocks (such as 1 to 7 in Figure 1*b*) immediately adjoining the study area are reported by *Cady* [1969, Table 3]. *Dallmeyer* [1979] recently obtained $^{40}Ar/^{39}Ar$ dates of about 385–395 m.y. for the regional metamorphism of the metasedimentary formations and 355–365 m.y. for the granitic intrusions located in south-central Maine. These ages are in conformance with the isotopic dates reported by *Naylor* [1971] and *Pajari et al.* [1974] and indicate that the duration of Acadian orogenic events covered a period of approximately 30 m.y. between 360 and 390 m.y.

Paleomagnetic Sampling and Measurements

A total of 235 oriented core samples were collected at 72 sites from metasediments (74 samples, 23 sites), hornfels (70

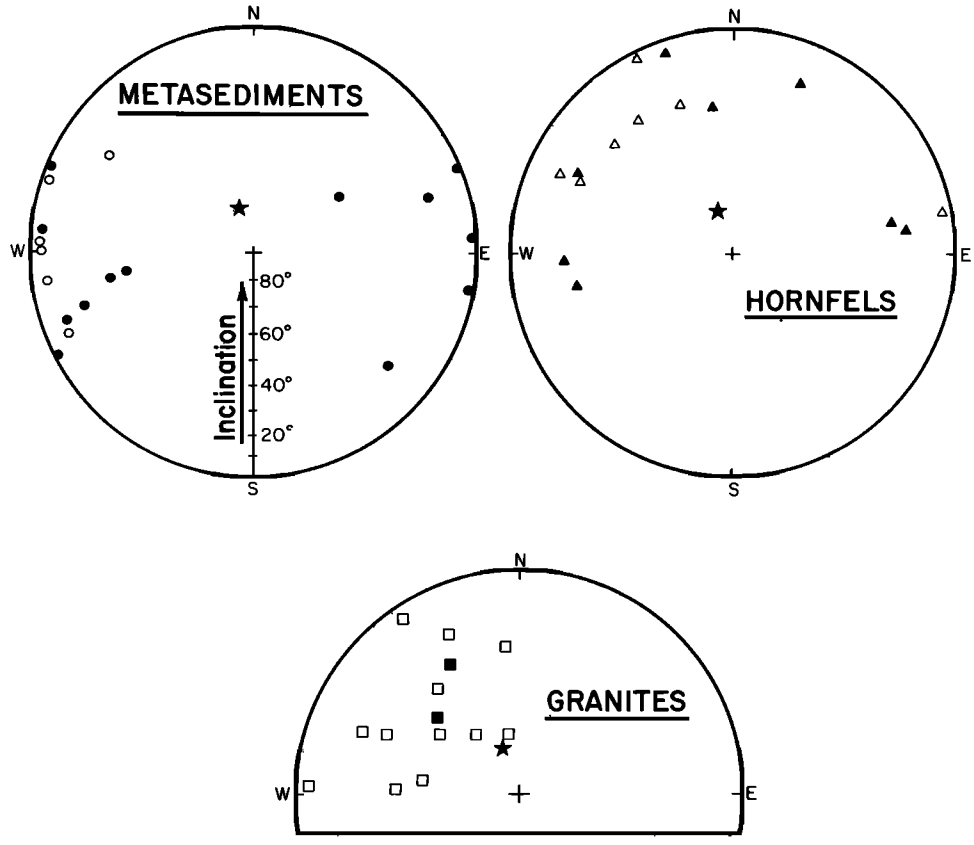


Fig. 2. Equal-area projections of the site-mean NRM directions plotted without tilt correction for each rock unit as indicated. Solid (open) symbols represent projections on lower (upper) hemisphere. The star represents the direction of the present earth's field at the sampling sites. The other symbols are as in Figure 1*c*.

samples, 16 sites), and granites (118 samples, 33 sites). The sampling was carried out with a portable diamond drill and orientations were measured with a Brunton compass. Since there are no suitable exposures, sampling could not be made at chilled margins associated with granite-hornfels contact and at the periphery of the contact aureole. With the intention of performing a paleomagnetic contact test [*Everitt and Clegg*, 1962], efforts were made to select the location of sites distance-wise with respect to the boundaries of the granite contact and the metamorphic aureole for hornfels and metasediments respectively. During field sampling we could not obtain cores from the layers of slate since they were found to be too fissile and friable to drill. Therefore the results obtained in the present study are mainly confined to siltstones (about 90%) since the majority of the sandstone samples collected were later found to be too weak to measure. The siltstones are often argillaceous; their color is variable but more generally pale to dark gray. Also during the course of the investigation, a total of 19 sites in granites (mainly aplitic and coarse grained type), 3 sites in metasediments (bordering the aureole), and 1 site in hornfels (bordering the intrusive contact) were discarded either because the samples were too weak to measure or unstable upon demagnetization. The locations of the remaining 49 sites used in the present study are indicated in Figure 1*c*.

The direction and intensity of natural remanent magnetization (NRM) were measured with a Schonstedt spinner magnetometer (DSM-1); stepwise demagnetization experiments were carried out using (1) Schonstedt thermal demagnetizer (TSD-1) in residual fields of less than 15 nT and (2) an alternating field (AF) demagnetizer with a three-axis tumbler in about 5 nT residual field [*Seguin*, 1975]. Stepwise AF and thermal procedures were employed up to 80 mT and 700°C, respectively, and *Zijderveld* [1967] vector diagrams were used in the subsequent analysis. For the NRM of each oriented sample, a minimum of one to a maximum of three specimens were used to obtain a sample-mean direction.

The NRM site-mean directions for the three rock types are shown in Figure 2 without correction for geological tilt. In general the directions are notably misaligned with the present earth's field (PEF) at the sampling sites which suggests the absence of a dominating overprint of recent origin. The majority of site-means in metasediments are predominantly southwesterly with shallow inclinations; granites are characterized by northwesterly, upward directions while those in hornfels are scattered northwest to northeast with both polarities. The NRM intensities vary from 10^{-3} to 10^{-1} A m^{-1} but the hornfels, in general, are more strongly magnetized than the granites and the metasediments.

Preliminary Demagnetization Results

For the preliminary treatment two samples from each site for each rock type were demagnetized by stepwise AF and thermal methods.

Metasediments (MS)

Upon AF demagnetization, the majority of Compton samples have lost 50% of their NRM intensity by 10 mT and shown a very erratic behavior at fields beyond 50 mT. Because of incomplete cleaning at higher fields, thermal treatment was preferred for metasediments in order to obtain

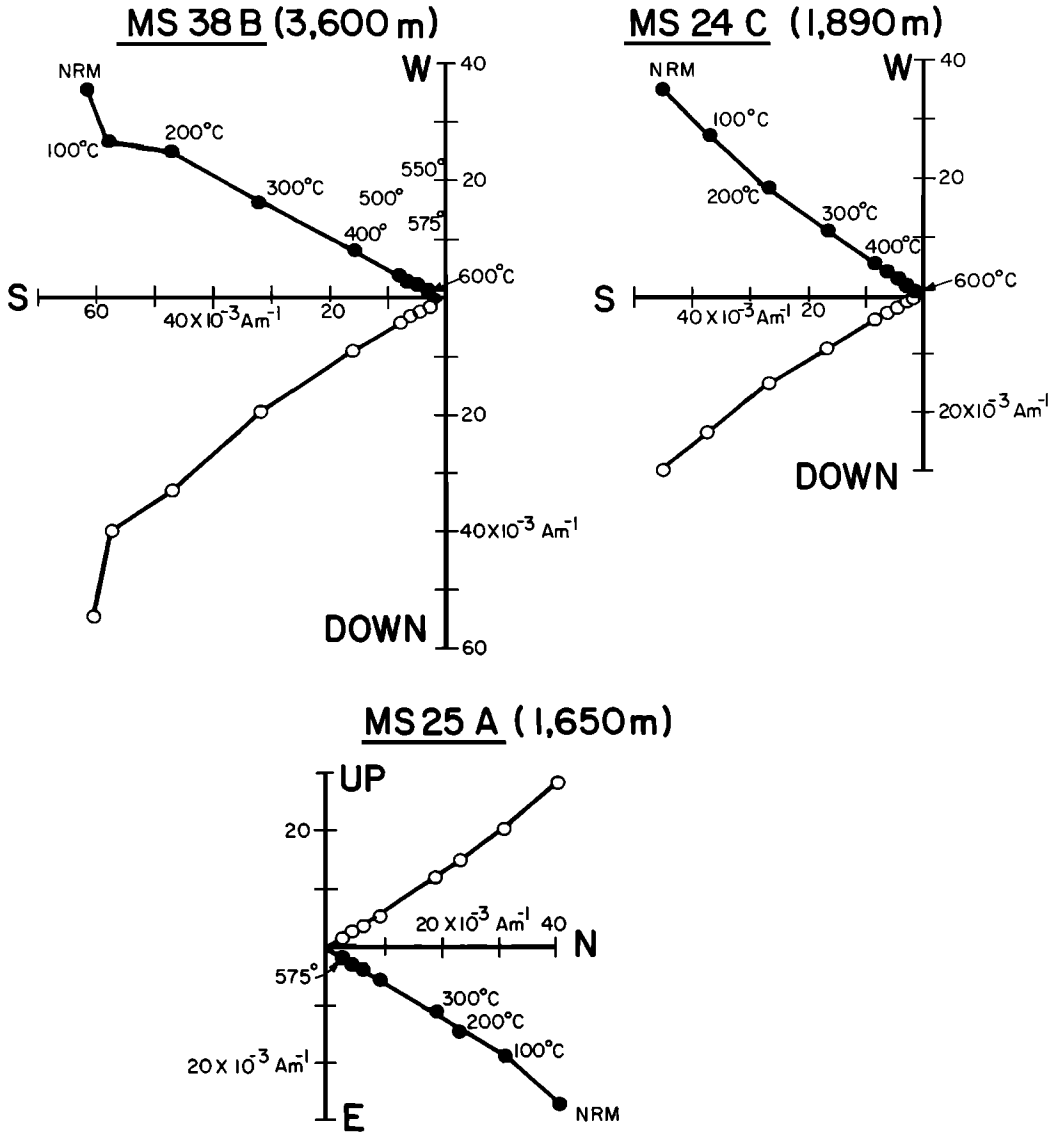


Fig. 3. Representative orthogonal thermal demagnetization diagrams [*Zijderveld*, 1967] of successive endpoints of magnetization vector for three specimens from metasediments (MS) taken from three different sites 38, 24, and 25 as indicated. The numbers written in brackets next to the specimen number denote the approximate distance in meters of the respective sampling sites from the nearby granite contact (Table 1). Solid (open) circles represent projections on horizontal (vertical) plane. Numbers next to solid circles represent the demagnetization steps in degrees centrigrade. Units indicated for axes are in 10^{-3} A m^{-1} ($=10^{-6}$ emu cm^{-3}).

a clear decay to the origin. The demagnetization behavior upon thermal treatment of the NRM of three representative samples from the Compton metasediments are characterized by a uniform linear decay of the magnetization vector (Figure 3). While a consistent southwesterly downward remanence is found to be present up to temperatures 500°–600°C in the majority of samples (MS38B, 24C; Figure 3), the presence of approximately antiparallel magnetization vectors (northeasterly, upward) are also observed in some samples (MS25A; Figure 3). The moderate values of unblocking temperatures (T_{uB}), 350°–500°C suggest magnetite as the most likely carrier of remanence for both southwesterly and northeasterly directions.

Hornfels (HF)

AF demagnetization was found to be more effective than thermal cleaning for rapid initial removal of randomly directed soft viscous remanence (≦5–10 mT) present at some sites in hornfels. Demagnetization of samples beyond 5–10 mT revealed a northwesterly upward remanence with a linear trajectory to the origin (HF39A, 73C; Figure 4). By 20 mT, most of the hornfels have lost half of their NRM intensity. The remaining intensity decays along linear trajectories (as expressed by three or more collinear points) obtained between 15–50 mT. It is therefore suggested that the remanence is carried by magnetite. Except in a few samples (HF16C; Figure 4), thermal demagnetization of hornfels produced erratic directions presumably due to chemical alteration of the magnetic minerals in the contact aureole or chemical changes generated during experiments and/or noise since the intensities are frequently too low (1 to 3 × 10^{-3} A m^{-1}) for measurement beyond 500°C. However the thermally obtained directions are not significantly different from those derived from companion specimens treated by AF.

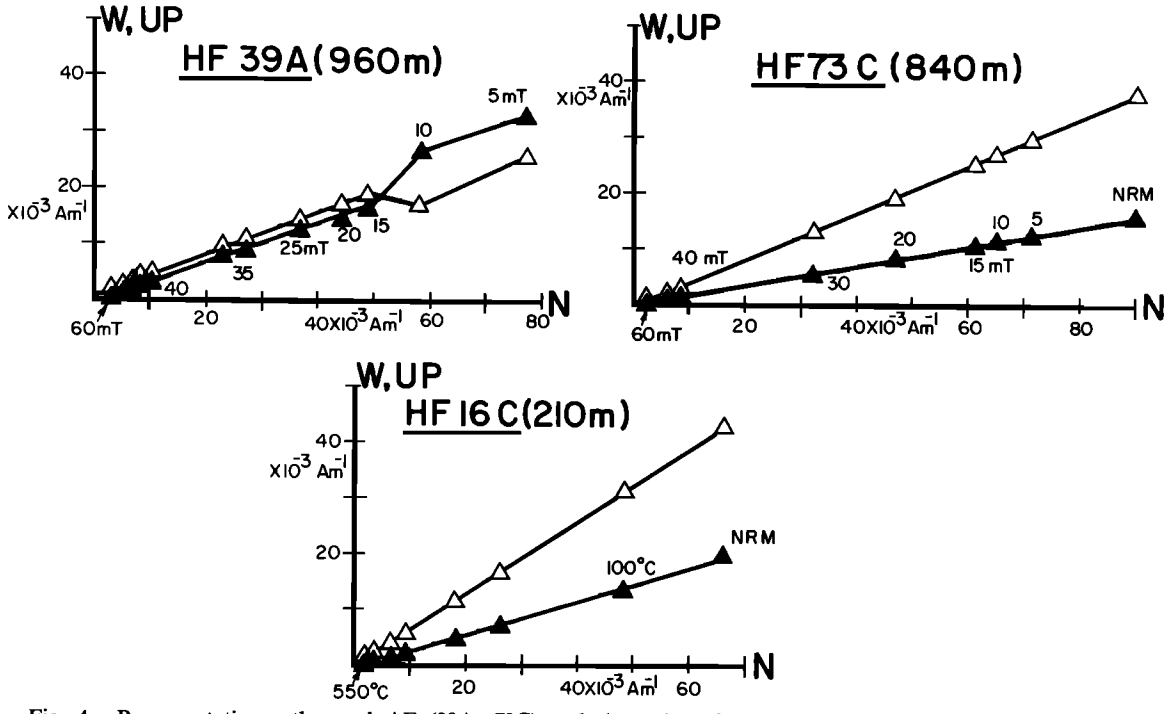


Fig. 4. Representative orthogonal AF (39A, 73C) and thermal (16C) demagnetization diagrams of successive endpoints of magnetization vector for three specimens from hornfels (HF). The other symbols are as in Figure 3.

Granites (G)

AF demagnetization of granites revealed univectorial northwesterly, upward directions of magnetization with minimum dispersion usually between 15 and 50 mT (G14-1; Figure 5*a*). The AF cleaned directions do not differ significantly from their NRM directions. In contrast, thermal treatment of granites is characterized by inconsistent directional (with a wider variation in declinations relative to AF treated results) and intensity (substantial increase or decrease) changes. However, in few well-behaved samples (G14-2; Figure 5*b*) the presence of a northwesterly upward

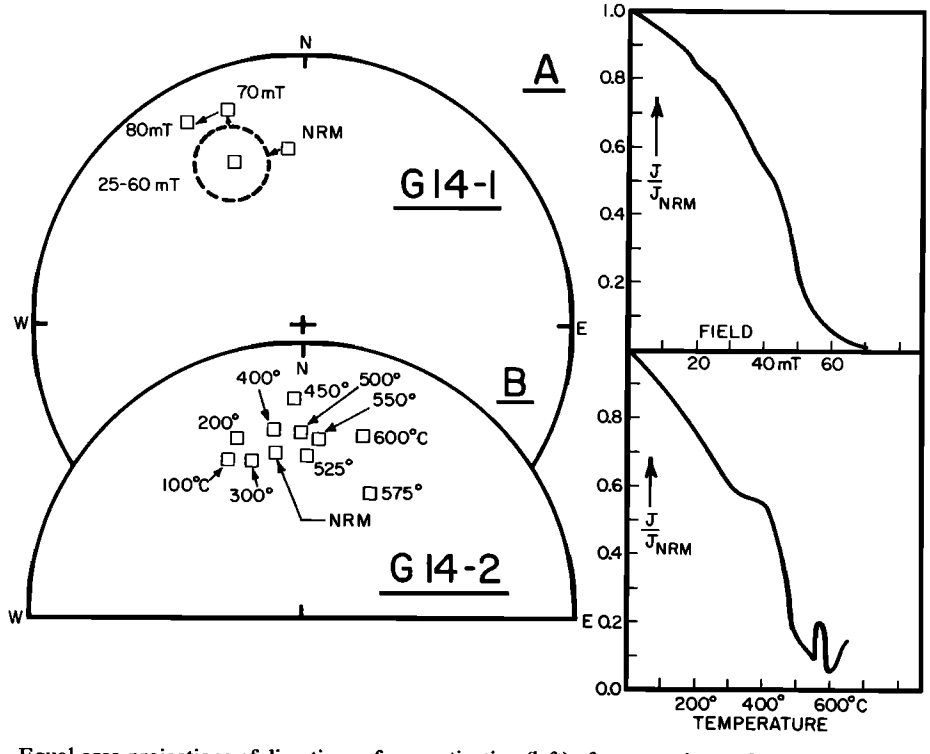


Fig. 5. Equal-area projections of directions of magnetization (left) of two specimens from the same granite sample (G14) after stepwise (*a*) AF and (*b*) thermal treatments. The numbers next to open squares represent the demagnetization steps in (*a*) mT and (*b*) °C as indicated. The respective normalized AF and thermal decay curves are shown on the right side.

TABLE 1. Compton Metasediments, Site-Mean Remanence Directions

Site	d, m	$N(n)$	NRM Before T.C. $D°, I°$	NRM Before T.C. k	After Thermal Treatment: Before T.C. $D°, I°$	After T.C. $D°, I°$	After T.C. k	After T.C. $\alpha_{95}°$	Range, °C*
					Metasediments				
21	4650	2 (2)	73, +20	...	30, +25	15, −9	...	...	300–400
38	3600	3 (6)	247, −11	20	235, −3	218, +22	110	11.9	200–550
22	3540	4 (4)	293, +2	13	190, +10	224, +25	120	8.2	300–500
23	3060	3 (5)	86, +11	7	28, +15	26, −13	89	13.1	300–400
37	2700	4 (7)	250, +11	15	250, +5	211, +3	70	11.0	300–500
33	2520	3 (3)	271, −5	16	250, +20	204, +28	229	8.2	400–500
68	2400	4 (6)	289, −22	8	240, +10	205, +28	170	7.1	300–500
28	2310	4 (4)	276, +66	5	230, +20	206, +5	38	15.2	400–550
34	2250	2 (2)	274, −44	...	250, +10	210, +31	...	...	400–500
29	2220	3 (5)	253, −22	24	200, +10	218, +20	130	11.4	400–550
43	2100	4 (4)	263, −7	38	203, −5	218, +24	240	5.9	400–550
24	1890	3 (6)	243, +11	12	186, +11	225, +29	125	10.8	400–550
25	1650	4 (6)	68, +22	14	60, −10	24, −25	78	10.4	300–500
46	1300	2 (4)	100, +22	...	38, +25	5, +7	...	...	400–550
47	1200	3 (5)	304, −24	20	25, +11	28, −14	150	10.1	400–550
Total 15 sites after treatment			Before T.C.: $D° = 221$, $I° = +2$; $k = 9$; $\alpha_{95}° = 16.8$						
			After T.C.: $D° = 209$, $I° = +20$; $k = 30$; $\alpha_{95}° = 7.2$						
			Pole Position (MS): N. lat. = 28°, E. long. = 77° dp, dm = 4°, 8°						
					~1100 m—Hybrid Zone				
27(MS)†	1140	4 (8)	262, +42	5	260, +40	183, +35	14	15.4	100–300
31(MS)†	1140	4 (6)	57, +53	14	10, +50	128, +37	47	13.6	100–400
32(MS)†	1140	3 (5)	130, +22	7	50, −25	323, −10	16	11.5	100–400
30(MS)†	1110	3 (4)	260, +35	60	190, +5	231, +50	75	8.2	200–500
45(HF)†	1050	3 (3)	82, +22	129	95, +10	235, +4	225	11.4	200–400

d, estimated approximate distance (in meters) of the sampling site from the nearby granite contact; N, number of samples averaged in Fisher statistics; n, number of specimens used in measurement, taken from N samples; $D°$, $I°$, mean declination and inclination of remanence at each site before and after tilt correction (T.C.); I is positive (negative) downward (upward); k, precision parameter; α_{95}, radius of 95% confidence circle (°); MS, metasediments; HF, hornfels; dp, dm, semi-axes of ovals of 95% confidence; pole MS plotted in Figure 5.

*Range of unblocking temperatures over which mean directions are calculated using Zijderveld trajectories; the hybrid zone mean directions are mostly based on minimum dispersion criterion in the temperature range shown.

†Rock type.

direction at low temperature (NRM −400°C) steps (similar to the one detected in AF treatment) is apparent but also accompanied by a high temperature (400°–550°C) northeasterly upward direction ($D \approx 40°$, $I \approx -15°$). Nevertheless, this high temperature direction (Figure 5*b*) is not commonly present in the companion specimens after AF treatment. Since we are not certain about the possible effect of chemical changes produced during heat treatment of granites, we limit our discussion only to the AF treated data.

Analysis of Paleomagnetic Directions

MS. For the majority of sites in the metasediments, detailed stepwise thermal treatment up to 600°C revealed no indication of secondary components as evidenced by linear decay of the magnetization vector toward the origin (Figure 3). Subsequent thermal treatment in three or four steps for the rest of the metasediment samples confirms the presence of a southwesterly direction with a $T_{\mu B}$ spectrum of 300°–500°C obtained in the pilot study. At five sites (21, 23, 25, 46, and 47), the presence of an approximately antiparallel direction (northeasterly) but with a similar $T_{\mu B}$ spectrum is also observed. Based on linear vector trajectories, respective site-mean values from sample-means for thermally treated directions before and after geological tilt correction are calculated (Table 1). The corresponding range of thermal treatment and the respective distance from the nearby granite contact for each site are also listed in Table 1.

However, four sites (27, 31, 32, and 30) near the south edge of the contact aureole and located approximately 1100 m from the intrusive contact (Figure 1*c*, Table 1) have shown different directions which are not characteristic of the metasediments. Hypothetically in the proximity of a contact aureole, a host rock is expected to record a component of magnetization acquired in the ambient field on slow cooling from the peak temperature attained in the aureole. Still, at three sites only one remanence directed northwest (27 and 31 approximately antiparallel to 32) which was commonly detected in hornfels is observed. At site 30, a southwesterly steep downward direction is obtained (after tilt correction) which is not significantly different from its NRM direction. The absence of a southwesterly (or northeasterly) direction characteristic of metasediments at the MS sites 27, 31, 32, 30 may be explained in terms of remagnetization acquired at higher temperatures probably reached at the time of contact metamorphism. Consequently, for statistical purposes these four sites are not included with the rest of the MS sites to obtain a mean direction for metasediments.

The site-mean (N = 15 sites) directions in the Compton Formation give a in situ formation mean at $D = 221°$, $I = +2°$, $k_1 = 9$, $\alpha_{95} = 16.8°$. After correction for bedding tilt (not corrected for gentle plunge) the corresponding direction moves to $D = 209°$, $I = +20°$, $k_2 = 30$, $\alpha_{95} = 7.2°$ (Table 1, Figure 6*a*). The northeasterly site-mean directions obtained from five sites (21, 23, 25, 46, 47) were inverted to obtain the formation-mean direction. The grouping of site-mean directions after bedding tilt correction is statistically significant

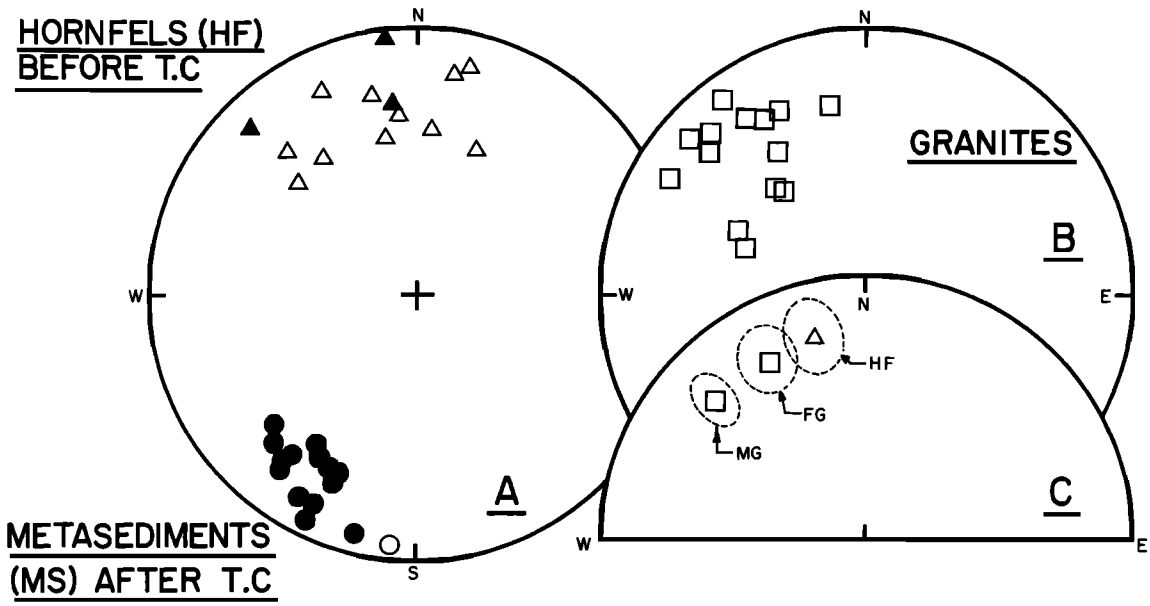


Fig. 6. Equal-area projections of site-mean directions of magnetization obtained: (*a*) for metasediments (MS) after thermal treatment and tilt correction (T.C.) (Table 1) and for hornfels (HF) after (AF) treatment but without T.C. (Table 2), (*b*) for granites (G) after AF treatment (Table 3). (*c*) Unit-mean remanence directions obtained for the medium-grained (MG), fine-grained (FG) granite sites (Table 3) and hornfels (HF, Table 2) along with the error ovals (α_{95}, Δ_{95}) (Tables 2, 3). Other symbols as in Figure 1*c* and Figure 2. Compare Figure 5 to Figure 6*b* and note the declination spread for granites.

since the value of $k_2/k_1 = 3.14$ for $N = 15$ is greater than the corresponding variance ratio (= 2.47) at 99% confidence limit [*McElhinny*, 1964]. Therefore the unfolded direction $D = 209°$, $I = +20°$ is considered to be a stable remanence retained in the tightly deformed sediments. A notable improvement (5 to 21 times) in within-site dispersion for each site relative to the corresponding NRM level and the univectorial nature of magnetization with $T_{\mu B}$ spectrum, 300°–550°C (Table 1), are two other significant features which attest the predeformational stability of this magnetization.

HF. The rest of the hornfels collection was demagnetized in three or more steps determined by the pilot AF study results. The majority of the site-mean directions in hornfels (Table 2, Figure 6*a*) are characterized by the following:

1. Marginal to substantial improvement (1 to 9 times) in within-site dispersion between 10- and 60-mT steps relative to the respective NRM site-means.

2. Four types of magnetization directions (Table 2) represented predominantly by a large spread in declinations (312° to 22°), a northwesterly upward direction *A* from six sites (20, 39, 49, 4, 78, 16), a northwesterly downward but shallow remanence *B* at three sites (73, 74, 60), a northeasterly upward direction *C* at two sites (55, 8), an approximately antiparallel (southeasterly, downward) direction to *A*, denoted as *D*, at three sites (19, 48, 6). For statistical purposes, the *D* directions were inverted to obtain a mean direction for the hornfels.

3. Scatter in site-mean directions (α_{95}, $\Delta_{95} = 39°, 41°$) after tilt correction (= simple rotation about strike). The large uncertainty in declinations is measured by $\Delta_{95} = \sin^{-1}(\sin \alpha_{95}/\cos I)$, i.e., the angle subtended between the mean declination and the tangent to the cone of confidence (α_{95}) from the center of an equal-area projection [*Kellogg and Reynolds*, 1978; *Beck*, 1980]. The scatter could mainly be attributed to uncertainty in plunging fold axes (particularly at sites near the periphery of the intrusive contact where the regional fold pattern is locally modified by the intrusion of granite).

4. Declination spread ($2\Delta_{95} = 32°$) before tilt correction could be explained in terms of the cooling history of the contact aureole. According to calculations by *Carslaw and Jaeger* [1959], temperatures of 400°C or more could be maintained for $\approx 10^4$ years at distances of $\approx$500 m from contacts of granitic plutons 1 or 2 km wide (such as Ste. Cécile–St. Sébastien granite) intrusive into dry sediments maintained at 100°C. Furthermore, the peak temperature attained in the aureole varies with increasing distance from the intrusive contact. Therefore, the declination spread could be attributed to secular variation or apparent polar wander during the slow acquisition of a thermal or chemical remanence. The declination discrepancies could also possibly arise from rotations (on a scale of a few meters to kilometers) undergone by the hornfels and granite rock units since the time of their emplacement, but field evidence in support of this is lacking.

Because of these uncertainties, the results of the hornfels should be taken with some reservation. However, it is noted that the grouping of site mean directions is better before than after correction for tilt; the ratio k_1/k_2, a measure of the statistical significance of a negative fold test equals 4, while a ratio of 2.6 is required for $N = 14$ at the 99% confidence level [*McElhinny*, 1964]. It would then appear that the hornfels remanence ($D = 346°$, $I = -22°$; Table 2) is of postdeformational origin.

Hybrid zone. A single site in hornfels (site 45) located easterly at about 1050 m from the granite, i.e., approximately at the edge of the aureole (Figure 1*c*) upon AF and thermal treatment resulted in a mean direction $D = 95°$, $I = +10°$ (Table 1) which is in disagreement with the other 14 sites. The corresponding mean direction after tilt correction, $D =$

TABLE 2. Compton Hornfels, Site-Mean Remanence Directions

Site	d, m	$N(n)$	NRM Before T.C. $D°, I°$	NRM Before T.C. k	After AF Treatment: Before T.C. $D°, I°$	Type of Remanence	After AF Treatment: After T.C. $D°, I°$	After T.C. k	After T.C. $\alpha_{95}°$	Range, mT*
20	1200	5 (8)	79, −4	8	335, −17	A	345, +47	69	9.3	10–50
19	1020	4 (4)	259, +30	11	137, +29	B	138, −42	43	16.5	20–50
39	960	5 (6)	29, +29	7	325, −28	A	61, −59	37	18.0	40–60
48	900	4 (4)	268, +26	16	193, +15	B	194, −17	28	15.4	20–40
73	840	4 (4)	79, +41	23	315, +14	C	329, −75	30	17.2	15–50
49	720	6 (12)	342, +7	16	349, −40	A	92, −46	110	6.5	15–60
4	720	5 (10)	22, +19	10	354, −33	A	69, −42	12	22.8	10–30
78	660	4 (6)	312, +29	25	312, −38	A	314, +59	390	4.7	15–50
6	390	4 (4)	194, +15	40	189, +18	B	219, +33	164	7.2	15–40
74	300	4 (4)	80, +28	14	352, +30	C	3, −41	14	25.2	5–25
60	300	5 (5)	334, −3	20	352, +2	C	43, +50	47	11.6	10–50
16	210	3 (6)	296, +26	18	347, −25	A	77, −35	270	7.6	10–60
55	60	4 (6)	297, +23	14	22, −41	D	89, −7	54	13.4	10–60
8	30	5 (8)	353, +35	40	4, −39	D	75, −34	81	8.4	10–60
Total 14 sites after treatment			Before T.C.: $D° = 346$, $I° = -22$; $R = 12.3257$; $k = 8$; $(\alpha_{95}, \Delta_{95})° = 15.3, 16.4$							
			After T.C.: $D° = 39$, $I° = -15$; $R = 7.5054$; $k = 2$; $(\alpha_{95}, \Delta_{95})° = 39.0, 40.7$							
			Pole position (HF)							
			Before T.C.: N. lat. = 32°, E. long. = 125°; dp, $dm = 8\frac{1}{2}°$, 16°							

Pole HF plotted in Figure 5; Δ_{95} is uncertainty associated with declination at 95% probability level ($\Delta_{95} = \sin^{-1}[\sin \alpha_{95}/\cos I]$). Other symbols as in Table 1.

*Range of coercivity values over which mean directions are calculated using Zijderveld trajectories.

235°, $I = +4°$ (preintrusion direction?) is not significantly different from average remanence derived from the metasediments. This is in contrast with the four sites (27, 31, 32, 40) in metasediments (but which are also located at the same distance, 1100 m, from the intrusive contact) which show directions similar to those observed in hornfels. We believe that the five sites (27, 31, 32, 40 in metasediments, 45, in hornfels) originated from a "hydrid zone" or a warmed zone (Table 1) in which the peak temperature attained during contact-metamorphism was substantially lower (say by 100°–200°C) than those in the contact aureole. Furthermore, the presence of pyrite and pyrrhotite was detected at MS sites 27, 31, 32 as evidenced from the examination of polished sections. This is in conformance with the low $T_{\mu B}$ range (100°–400°C, Table 1) observed at these sites. *Kelly* [1975] reported geochemical evidence for the existence of pyrrhotite associated with molybdenum mineralization in fractures and openings at the intrusive contact and surrounding region. The mineralization is interpreted by Kelly as a product of metasomatism. Generally, the presence of pyrrhotite gives rise to self-reversal and anisotropy of susceptibility by virtue of which it can become magnetized in a different direction from that of the applied field [*Fuller*, 1963]. We believe therefore that the remagnetization of metasediments and hornfels in the hydrid zone was the combined result of granitic intrusion and simultaneous introduction of pyrrhotite by metasomatic action.

Granites. Following the pilot AF results, the rest of the granite collection was bulk demagnetized in three or four steps between 25 and 50 mT and site-mean directions were computed (Table 3, Figure 6) based on minimum dispersion criterion. Following the classification of granites by *Kelly* [1975] the sampling sites are grouped to obtain a mean remanence direction for the fine (FG) and medium (MG) grained granites. After treatment, the presence of a univectorial northwesterly directed magnetization similar to the one derived from hornfels (*A*-type remanence, Table 2) is obtained. This is characterized by marginal improvement (1 to 10 times) in within and between-site dispersions as well as directional stability of remanence relative to NRM directions (Table 3). While the values of declination uncertainty obtained for FG ($\Delta_{95} = 18°$) and MG ($\Delta_{95} = 13°$) units are comparable to that of hornfels ($\Delta_{95} = 16°$, Table 2), except for *A*-type remanence, no *B*, *C* and *D* type remanences are present in the granites. While the site-mean directions before and after AF treatment for the FG sites do not differ very much from each other, the cleaned directions for MG sites have a pronounced deviation relative to their NRM directions. Mean remanence directions corresponding to a total of 14 sites in granites (irrespective of classification into FG and MG units) are also computed for NRM and after AF treatment (Table 3, Figure 6*b*). The plot of HF, FG and MG site-mean directions along with their (α_{95}, Δ_{95}) ovals (Tables 2, 3) in Figure 6*c* indicate that FG (14, 18) and HF (15, 16) ovals overlap with each other while the FG and MG (12, 13) unit-mean remanences are significantly different. This implies that FG and MG units were emplaced at two different periods of time and that FG and HF magnetizations were contemporaneous. The predominant occurrence of large patches of fine-grained granite (such as sites 11 to 15, Figure 1*c*) within the medium-grained granite in the central part of the stock prompted *Kelly* [1975] to suggest that the aplitic and fine-grained phases were younger than the medium-grained phase. This may then indicate that hornfels were mainly produced during the intrusion of the FG unit. However, the classification scheme of the granite body into three easily recognizable phases and the composite nature of the stock requires a thorough structural and petrographic analysis. In the absence of such a detailed study, it is premature to infer from Figure 6*c* (1) a sequence in the time of emplacement for MG and FG magnetizations and (2) contemporaneity of FG and HF magnetizations. Most likely the declination-spread in FG and MG could also be the record of paleosecular variation. However, the comparison of granite

TABLE 3. Ste. Cécile–St. Sébastien Granite, Site-Mean Remanence Directions

Site	N(n)	NRM		AF (25–50 mT)		
		D°, I°	k	D°, I°	k	α_{95}°
		Fine-Grained Granites				
11	5 (10)	324, −63	12	321, −50	30	13.5
12	5 (8)	307, −53	7	318, −48	23	15.6
13	4 (8)	351, −68	14	335, −26	144	8.2
14	5 (10)	355, −34	19	328, −38	198	5.4
15	3 (6)	322, −40	44	350, −30	137	10.8
	5 sites	332, −53	18	332, −39	31	(α_{95}, Δ_{95}) = 13.9°, 18.0°
		Pole position (FG): N. lat. = 18°, E. long. = 137°; *dp*, *dm* = 10°, 17°				
		Medium-Grained Granites				
51	4 (8)	273, −44	11	290, −50	26	14.4
52	5 (10)	295, −36	6	312, −22	44	18.6
53	4 (6)	278, −54	5	295, −46	43	13.2
59	3 (6)	292, −25	12	311, −13	62	15.5
69	5 (5)	326, −5	7	330, −25	40	14.6
70	5 (5)	337, −23	7	316, −18	61	9.8
72	5 (10)	312, +49	16	324, −10	24	15.6
76	4 (6)	333, +35	20	326, −22	41	14.2
77	4 (8)	273, −6	16	300, −15	64	12.2
	9 sites	303, −15	4	313, −25	19	(α_{95}, Δ_{95}) = 12.0°, 13.4°
		Pole position (MG): N. lat. = 18°, E. long. = 158°; *dp*, *dm* = 7°, 13°				
Total	14 sites (all granites)	312, −31	4	319, −30	17	(α_{95}, Δ_{95}) = 9.9°, 11.4°
		Pole position (G): N. lat. = 18°, E. long. = 150°; *dp*, *dm* = 6°, 11°				

Pole positions FG, MG, and G plotted in Figure 5. Other symbols as in Tables 1 and 2.

(Table 3, Figure 6*b*) and the hornfels (Table 2, Figure 6*a*) mean-remanence directions suggest only that they are indistinguishable from each other.

Magnetic Mineralogy

Magnetic mineralogy of the three rock types was identified by polished section examination of twelve representative samples with a Reichert reflecting microscope (magnification 150 to 700). The metasediments contain a small quantity of fine-grained magnetite highly dispersed in the ground mass; very little hematite has been seen. In the hornfels, very fine rounded grains of magnetite are present. At sites around the edge of the contact aureole, pyrite and pyrrhotite are the additional opaque minerals commonly observed both in metasediments and hornfels. In granites, magnetite is present in large rounded grains clustering around rutile and occasionally with ilmenite. The most important volume of opaque minerals common in all the three rock types is magnetite.

We also measured the acquisition of isothermal remanent magnetization (IRM) and remanence-coercivity (H_{cr}) spectra at room temperature in fields up to 1.4 T to infer the magnetic mineralogy of the three rock types (four representative samples for each rock type). Irrespective of the rock type, all the samples reached saturation within an applied field of 0.1–0.2 T and characterized by the values of H_{cr} = 0.01–0.04 T confirming the presence of magnetite [*Dunlop*, 1972] as observed from optical examination. While magnetite is observed to be the common carrier of remanence in the three rock types, the effective method of demagnetization is found to be thermal for MS and AF for HF and G units. We preferred AF treatment to thermal for HF and G units to prevent chemical changes produced during heating, possibly due to the presence of minor amounts of pyrite and pyrrhotite in these rock units. In the case of MS, we made use of thermal demagnetization because the remanence directions were found to be erratic beyond 40 mT during AF treatment.

Paleomagnetic Poles

Paleomagnetic poles MS, HF and G (Tables 1, 2, 3; Figure 7) are computed respectively for the prefolding remanence (D = 209°, I = +20°) retained in the metasediments, the postdeformation remanence (D = 346°, I = +22°) by hornfels and the mean-remanence (D = 319°, I = +30°) for granites. The corresponding poles are also shown separately for the fine (FG) and medium (MG) grained granite sites (Table 3, Figure 7). Available regional geological syntheses on the Gaspé-Connecticut Valley synclinorium (Table 1*b*) place the age of folding and metamorphism (sub-greenschist facies) at about Middle Devonian time (375–385 m.y.) and the time of intrusive activity at about 360–370 m.y. These age estimates are in broad agreement with the early Middle to late Early Devonian age for the Compton metasediments based on a preliminary study of fossil plants in the area [*Kelly*, 1975] and the K-Ar date of about 362 m.y. as an age of intrusion for the Ste. Cécile–St. Sébastien granite [*Lowdon*, 1960]. Assuming that the Middle (375–385 m.y.) and Late (365–375 m.y.) Devonian ages represent respectively the ages of folding and intrusion during Acadian orogeny, we infer that the corresponding age for the pole MS derived from prefolding remanence in metasediments to be at least late Early or early Middle Devonian. While we are not certain (in the absence of stability criteria and strong evidence for a two or three phase intrusive history of the granitic stock) regarding the relative age difference between the FG and MG (or G) unit-mean remanences, the granite poles may correspond to a general Middle-Late Devonian age. Upon consideration of associated error oval (*dp*, *dm*) values (Tables 1, 2, 3), the postintrusive HF pole does not differ significantly from the granite pole FG which may be taken as an evidence that the two poles FG and HF are probably contemporaneous.

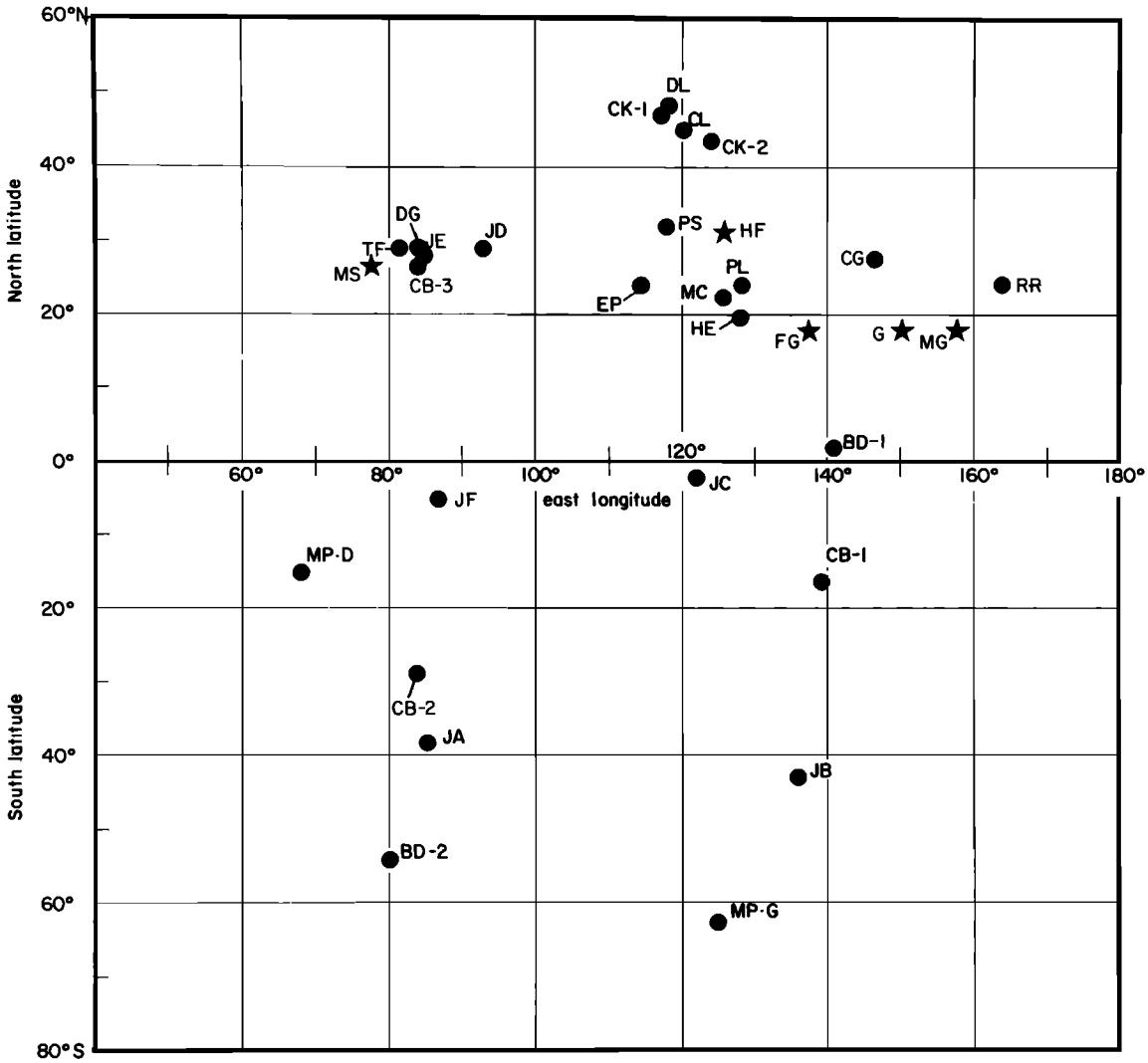


Fig. 7. Paleomagnetic poles obtained from Devonian rock units from northern Appalachians. DG, Dockendorff group volcanics [*Brown and Kelly*, 1980]; TF, Traveler Felsite [*Spariosu and Kent*, 1980]; JE, JF Mascarene group felspathic greywackes and siltstones [*Roy and Anderson*, 1981]; JA, JD, St. George's pluton [*Roy et al.*, 1979]; CB-1, CB-2, CB-3, Cape Breton granites [*Rao et al.*, 1981]; CK-1, Catskill redbeds [*Kent and Opdyke*, 1978]; CK-2, Catskill redbeds [*Van der Voo et al.*, 1979]; CL, Columbus limestone [*Martin*, 1975]; DL, Delaware limestone [*Martin*, 1975]; PS, Perry sediments [*Robertson et al.*, 1968]; PL, Perry Lavas [*Black*, 1964]; MC, Mafic Complex [*Schutts et al.*, 1976]; EP, Eastport Formation [*Kent and Opdyke*, 1980]; H, Hersey Formation [*Kent and Opdyke* 1980]; CG, Clam Bank group sediments [*Black*, 1964]; RR, Raisin River dolomite [*Martin*, 1975]; JC, Mascarene Group intrusives [*Roy et al.*, 1979]; JB, St. Stephen gabbro [*Roy et al.*, 1979]; MP-D, MP-G, Mt. Peyton diorite and granite [*Lapointe*, 1979]; BD-1, BD-2, Bonavista Bay dikes [*Murthy and Rao*, 1978]. Poles represented by stars are obtained in the present study from MS (metasediments), HF (hornfels), G (all granites), FG (fine) and MG (medium) grained granite units. Two poles: one from Belchertown pluton (tectonic rotation; *Ashwal and Hargraves* [1977]) and the other from Onondaga limestone (Permian remagnetization; *Kent* [1979]) are not plotted. Another pole derived from Perry sediments [*Black*, 1964] is not plotted for clarity purpose. For the detailed description of the geographic location of different sampling regions, and the tectonic features pertinent to the above paleomagnetic studies, the reader is referred to *Kent and Opdyke* [1978], *Roy and Anderson* [1981], *Rao et al.* [1981], *Roy* [1982], *Kent and Opdyke* [1982].

A comparison of the available Siluro-Devonian poles from Appalachians with the data obtained in the present study (Figure 7) reveals that the pole MS is in excellent agreement with other poles derived from nearby regions, such as New Brunswick, JE (felspathic greywackes—Waweig age [*Roy and Anderson*, 1981] and JD (St. George gabbro—390 m.y. [*Roy et al.*, 1979]); Maine, DG (Dockendorff group volcanics—Middle Devonian [*Brown and Kelly*, 1980]) and TF (Traveler Felsite—Lower Devonian [*Spariosu and Kent*, 1980]); and Nova Scotia, CB-3 (Cape Breton granites-inferred age ~400 m.y. [*Rao et al.*, 1981]). Consistency of the five poles (MS, JE, JD, TF, CB-3—mainly Lower Devonian) derived mostly from the southeastern margin of the presumed proto-Atlantic [*Wilson*, 1966] with a grouping located approximately at 30°N, 80°–90°E is tectonically significant [*Rao et al.*, 1981]). The significance of this result cannot be ascertained in the absence of contemporaneous poles derived from the cratonic margin of North America, but attention must be given to the fact that most of these poles are derived from multiphase remanence directions. Another exception to the above grouping of poles is the younger DG-pole derived from post-Acadian remanence direction. It is

difficult to explain this discrepancy in view of the limited nature of the fold test, in which only one site is located on the eastern part of the Chapman syncline while the other three sites form the western limb [*Brown*, 1979; *Brown and Kelly*, 1980].

However, the grouping of predominantly Early Devonian poles is in striking contrast with the other Middle-Late Devonian poles reported from northern Appalachians (Figure 7). Since the sampling area in the present study (Figure 1*c*) is located north of the proposed shear zone [*Kent and Opdyke*, 1978], the paleopoles HG and FG are supposed to be closer to the poles derived from cratonic margin of North America (cratonic poles CK-1, CK-2, CL, DL). Instead the HF and FG poles fall in the general area of Devonian poles reported for Acadia displaced terrain (PS, PL, MC). However, the error ovals (*dp*, *dm*) associated with the poles HF (8½°, 16°) and FG (10°, 17°) place the corresponding pole positions relatively close to the general area of so-called cratonic poles. But the structural [*St. Julien and Hubert*, 1975] and seismic [*Seguin*, 1982] evidence demonstrates the Gaspé–Connecticut Valley synclinorium as a nappe or thrust transported a few kilometers northwest toward the Internal domain (Figure 1*b*) of Quebec Applachians perhaps sometime during Acadian orogeny. Moreover, *Roy and Anderson* [1981] and *Rao et al.* [1981] question the significance of the paleolatitude difference between the poles derived from the Acadia displaced terrain of *Kent and Opdyke* [1978] and those from the cratonic margin of North America. Therefore we cannot make use of the poles HG and FG to test the validity of *Kent and Opdyke*'s [1978] hypothesis.

Furthermore the two other poles EP and HE which are older in age (upper most Silurian-Lower Devonian) and derived from eastern Maine [*Kent and Opdyke*, 1980] do not differ significantly from the poles HF, FG. Neither this agreement between the older (EP, HE) and the younger (HF, FG) poles nor the disagreement between almost contemporaneous groups of poles (DG, TF and HE, EP) derived from areas of similar tectonic history in Maine Appalachians, can be explained in terms of a geologically reasonable [*Ludman*, 1981] and paleomagnetically acceptable hypothesis. The paleomagnetic pole positions obtained for hornfels (HF) and granites (FG, MG, G) spread over 120° to 160°E in longitude. Such a difference cannot be explained in terms of a significant difference in the respective geological age between granites and hornfels. However, the respective mean-remanence directions for the HF and FG are not significantly different (Figure 6*c*) and therefore we believe that they are nearer in time, probably Middle to Late Devonian in age. Since there are difficulties in establishing the rock and remanence age for the granite (two or three phase intrusive history) and hornfels (declination discrepancy), we regard only the metasediment pole MS (28°N, 77°E) with probable late Early or early Middle Devonian age as reliable for apparent polar path construction.

CONCLUSIONS

A paleomagnetic study of three Devonian rock units (metasediments of the Compton Formation, Ste. Cécile–St. Sébastien granite intrusive and the surrounding hornfels) from 48 sites exposed in the Ste. Cécile–St. Sébastien region of the Gaspé–Connecticut Valley synclinorium in Quebec Appalachians, revealed the presence of two predominant types of stable magnetizations. Detailed thermal treatment of the tightly folded metasediments from 15 sites yielded a univectorial mean remanence at $D = 209°$, $I = +20°$, $k = 30$, $\alpha_{95} = 7.2°$ with a range in $T_{\mu B} = 300°$–$550°C$ carried probably by magnetite. The predeformational origin of this remanence is established by a fold test which is positive at 99% confidence level, as well as antipodal normal and reverse polarity directions. The corresponding paleomagnetic pole MS, falls at 28°N, 77°E (*dp*, *dm* = 6°, 11°) in agreement with the other poles, mostly Early Devonian in age and derived from the southeastern margin of the presumed proto-Atlantic ocean. In contrast, AF demagnetization of granites collected from 14 sites resulted in a univectorial mean remanence direction carried by magnetite at $D = 319°$, $I = -30$; $k = 17$, $\alpha_{95} = 10°$ with a corresponding pole, G, at 18°N, 150°E (*dp*, *dm* = 6°, 11°). Admitting the absence of a convincing evidence for a two or three phase intrusive history and composite nature of the Ste. Cécile–St. Sébastien granite, paleomagnetic poles corresponding to the fine (FG) and medium (MG) grained granitic phases are also calculated. They fall respectively at 18°N, 137°E (*dp*, *dm* = 10°, 17°) and 18°N, 158°E (*dp*, *dm* = 7°, 13°). While the reported single K-Ar result places the age of the pluton at 362 m.y., it is difficult paleomagnetically to ascertain the age of this remanence. AF demagnetization of hornfels collected from 14 sites revealed four types (*A*, *B*, *C*, *D*) of remanences characterized by large declination spread (≃60°) but with a mean remanence direction carried by magnetite at $D = 346°$, $I = -22°$; $k = 8$, $\alpha_{95} = 15°$ and which probably originated during postintrusive times. The corresponding pole, HF, falls at 32°N, 125°E (*dp*, *dm* = $8\frac{1}{2}°$, 16°). Since the confidence ovals of the FG and HF remanences overlap there is evidence that they are indistinguishable from each other. This may mean that the FG and HF remanences are proximal in time and probably originated during the Middle to Late Devonian period. Anomalous site-mean directions obtained from metasediments and hornfels located approximately 1100 m, from the intrusive contact (hybrid zone) are not considered for the calculation of MS and HF poles.

Since the HF and G remanences suffer from declination discrepancies and consequently spread over 40° in longitude in their corresponding pole positions, it is premature to make use of these data to confirm or negate possibility of Middle Paleozoic transcurrent motion postulated for the Northern Appalachians. We regard only the pole MS with probable late Early or early Middle Devonian age as reliable for the apparent polar path construction.

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